

Robotic Arm Simulation Project

Objective

Design and simulate a robotic arm capable of performing a basic **pick-and-place operation** using software such as:

- AutoCAD
 - SolidWorks
 - Fusion 360
 - MATLAB
 - Blender
-

1. Introduction

A robotic arm is a programmable mechanical device designed to imitate the movement of a human arm. It is widely used in manufacturing, material handling, welding, painting, and assembly operations.

The robotic arm used in this project performs:

- Object picking
 - Object lifting
 - Object placement at another location
-

2. Degrees of Freedom (DOF)

The **Degree of Freedom (DOF)** of a robotic arm refers to the number of independent movements or motions the robot can perform.

Each rotational or linear joint contributes one DOF.

Example:

A robotic arm with:

- Base rotation
- Shoulder movement
- Elbow movement
- Wrist movement

has **4 DOF**.

Types of Movements in DOF

Joint	Motion Type	Function
Base	Rotation	Rotates arm left/right
Shoulder	Angular	Raises/lowers arm
Elbow	Angular	Extends/retracts arm
Wrist	Rotation	Adjusts orientation
Gripper	Open/Close	Picks objects

3. Proposed Robotic Arm Design

Type of Robot

4-DOF Pick-and-Place Robotic Arm

Main Components

- 1. Base
 - 2. Shoulder Link
 - 3. Elbow Link
 - 4. Wrist Joint
 - 5. Gripper
 - 6. Servo Motors/Actuators
-

4. 2D/3D Model Description

Dimensions (Example)

Component	Length
Base Height	80 mm
Shoulder Link	150 mm

Component	Length
Elbow Link	120 mm
Wrist Link	70 mm
Gripper	50 mm

5. Design Procedure in SolidWorks/AutoCAD

Step 1: Create Individual Parts

Design:

- Base
- Arm links
- Joints
- Gripper

Use:

- Extrude
 - Revolve
 - Fillet
 - Hole features
-

Step 2: Assemble Components

Use assembly constraints:

- Revolute joints
 - Fixed mates
 - Coincident mates
-

Step 3: Define Motion

Assign:

- Motor rotation angles

- Joint limits
 - Motion sequence
-

6. Motion Simulation (Pick-and-Place)

Operation Steps

Step 1: Initial Position

Arm remains at home position.

Step 2: Rotate Base

Base rotates toward object.

Step 3: Lower Arm

Shoulder and elbow joints move downward.

Step 4: Grip Object

Gripper closes around object.

Step 5: Lift Object

Arm raises object upward.

Step 6: Move to Destination

Base rotates toward destination.

Step 7: Release Object

Gripper opens and releases object.

7. Joint Motion Angles (Example)

Joint	Initial Angle	Final Angle
Base	0°	90°
Shoulder	30°	60°
Elbow	45°	20°
Wrist	0°	45°

8. Kinematic Representation

The robotic arm movement can be represented using coordinate transformations and joint rotations.

For a planar robotic arm:

$$x = l_1 \cos \theta_1 + l_2 \cos(\theta_1 + \theta_2)$$

$$y = l_1 \sin \theta_1 + l_2 \sin(\theta_1 + \theta_2)$$

Where:

- (l_1, l_2) = link lengths
 - (θ_1, θ_2) = joint angles
-

9. Applications of Robotic Arms

Industrial Applications

- Assembly lines
- Welding
- Material handling
- Packaging

Medical Applications

- Surgical robots
- Rehabilitation systems

Domestic Applications

- Smart home automation
 - Service robots
-

10. Advantages

- High precision
- Fast operation
- Reduced human effort
- Improved safety

- Repeatability
-

11. Limitations

- High initial cost
 - Complex programming
 - Maintenance requirements
 - Limited flexibility without AI
-

12. Conclusion

The robotic arm simulation demonstrates the working principle of industrial robotic manipulators used in automation systems. By designing a 4-DOF robotic arm and simulating a pick-and-place operation, the project helps understand robot kinematics, motion control, and industrial automation concepts.

Simulation software such as SolidWorks and AutoCAD provides an effective platform for modeling and testing robotic systems before physical implementation.